# Sebastian Horta Donneys

New York | (516) 974-2139 | sh4506@columbia.edu | linkedin.com/in/sebastianhorta03 | github.com/SebasHorta | Portfolio

## Education

**Columbia University (SEAS)** — New York, NY
*B.S. Computer Science, Minor: Electrical Engineering* — *Aug 2023 – May 2026*

- Coursework: Computer Networks, NLP, Adv. C++ Systems Programming, Competitive Programming, Artificial Intelligence, Data Structures & Algorithms, Systems Programming, CS Theory, UI Design, Computer Systems
- Resident Adviser: Organized 10+ community events for 700+ residents, leveraging cross-functional teamwork

## Experience

**Machine Learning Engineer — Columbia University Robotics Lab** — *2025*
*New York, NY*

- Built **ML** pipeline in **Python** to detect gait events (heel strikes/toe offs) from VR motion-tracker data
- Engineered unsupervised **KMeans** clustering to classify VR trackers by body segment and foot side
- Developed **cross-correlation-based synchronization** between VR tracker streams and MAT ground truth
- Trained **Random Forest** classifier, correctly detecting ∼**85%** of gait events from VR tracker data

**Full Stack Software Developer — Freelance** — *2025 – 2026*
*Remote*

**Web Archiver Tool | React + FastAPI** — 2025

- Built offline web archiver with structured storage and versioning for **research** and offline site analysis
- Developed **recursive crawler** using **requests + BeautifulSoup** with URL rewriting for irregular link structures
- Created **React** dashboard with real-time progress tracking and iframe previews

**Client Business Website (Live Site) | Next.js + TypeScript + Tailwind** — 2026

## Projects

**Graph-Scoped Context Server for LLM Coding Agents | Python + MCP** — 2026

- Built an **MCP server** that scopes AI coding-agent context using dependency graphs and git-diff filtering
- Built a **benchmark harness** with 30+ faults to compare graph-based and baseline retrieval
- Implemented **dependency-graph traversal**, automated bug injection, and live MCP tool integration
- Designed an evaluation pipeline measuring **token efficiency** and fault-localization accuracy

**Allowance | Python** — 2026

- Designed a **P2P blockchain** enforcing AI agent spending limits via **ECDSA**-signed transactions and **PoW**
- Implemented **Merkle tree** transaction integrity and fork resolution to maintain consensus across 4+ nodes
- Built a peer discovery tracker with heartbeat monitoring, persistent **TCP** connections, and thread-safe state
- Validated correctness with **106 tests** covering transaction signing, overspend rejection, and tamper resistance

**C++ Generic Database & Build Tool | C++** — 2026

- Built a reusable **C++ database library** using class templates to store and retrieve any data type from disk
- Implemented a **build automation tool** supporting Makefile parsing, variable substitution, and implicit rules
- Resolved build dependencies recursively with **cycle detection**, only rebuilding targets whose source files changed
- Accelerated startup by caching parsed build rules to disk, skipping re-parsing when the Makefile is unchanged

## Skills

**Languages & Frameworks**: Python, Java, C/C++, JavaScript/TypeScript, HTML/CSS, React, Next.js, Tailwind
**Backend & AI/ML**: FastAPI, Flask, Node.js, REST APIs, SQL, scikit-learn, pytest, OpenCV, LLMs, MCP
**Tools**: Git, Unix/Linux, CI/CD, Figma

## Leadership & Activities

**SHPE at Columbia** – Board Member – Planned cultural and career events, resulting in increased presence on campus
**IEEE, ColorStack, CU ADI, CU BJJ, Sabor Dance** – Member – Club events and community building
**Collegiate Science & Technology Entry Program (CSTEP)** – Member – STEM outreach and program initiatives